Adaptive Dynamic Programming-Based Fixed-Time Optimal Control for Wheeled Mobile Robot

Chen Wang, Haoran Zhan, Qing Guo, *Senior Member, IEEE*, and Tieshan Li, *Senior Member, IEEE*

***Abstract*—In this study, the adaptive dynamic programming (ADP)-based fixed-time optimal trajectory tracking control is investigated for wheeled mobile robots. An ADP-based fixed-time optimal tracking controller is developed based on the critic-only neural network ADP technique, which guarantees the robot track the desired trajectory in fixed time. Firstly, to address the solution difficulty of the Hamilton-Jacobi-Bellman (HJB) equation, a critic neural network is used to estimate the cost function. Meanwhile, a weight update law is designed by using the adaptive control technique, which not only removes the persistent or finite excitation condition, but also enables the fixed-time convergence of the weight estimation error. By using the proposed controller, all error variables can converge to a neighborhood of zero in fixed time. Finally, both simulations and physical experiments indicate that the proposed ADP-based fixed-time optimal controller has a faster convergence rate compared to the two comparison controllers.**

***Index Terms*—Wheeled mobile robots, adaptive dynamic programming (ADP), fixed-time optimal control, critic neural networks.**

I. Introduction

THE wheeled mobile robot, with its superior mobility and flexibility, is gradually becoming an indispensable tool in various fields of modern society. Its efficient fulfillment of tasks requires the precise and stable support of a powerful control algorithm. However, the underactuated and nonlinear characteristics of the wheeled mobile robot introduce numerous complexities and challenges to the development of the control algorithms [1].

Trajectory tracking control, aimed at guiding wheeled mobile robots to follow a preset reference signal, has been a significant focus in the control community. Over the past decade, many control algorithms have been developed to address the trajectory tracking problem of wheeled mobile robots [2], [3], [4], [5], [6], [7]. Considering the possibility of slippage when the robot moves in slopes or uneven terrain, the authors in [2] and [3] proposed adaptive control algorithms with an adaptive compensation mechanism. To solve the difficulties posed by lateral and longitudinal slippage, Chen et al. [4] proposed an improved linear active disturbance rejection control algorithm for a wheeled robot with six wheels. For unicycle mobile robots with perturbation parameters, two sliding-mode control algorithms were proposed in [5] and [6] to achieve trajectory tracking. Additionally, considering wheeled mobile robots with unknown disturbances and unmeasurable velocities, a robust tracking control algorithm based on an extended observer was introduced in [7]. It is important to note that the above research mainly focuses on handling unknown disturbances, slippage, unmeasurable velocity information, and model parameter perturbations. However, these approaches only lead to asymptotic convergence of the system, meaning that the tracking error theoretically takes an infinite amount of time to converge to the equilibrium point or a neighborhood near it.

In recent years, a new control method, finite-time control, which ensures system convergence within a finite time, has gained popularity among scholars [8], [9], [10]. In [8], a fuzzy finite-time control method was proposed for nonlinear systems with actuator failures. In [9], a fast finite-time control scheme was proposed for nonlinear systems with full-state constraints to improve the convergence speed when the energy function value is relatively small. Later, this approach was extended to consensus control of nonlinear multiagent systems [10]. In addition, the finite-time control methods for wheeled mobile robots can be found in [11] and [12]. Li et al. [11] proposed a finite-time control method for time-varying systems and applied it to the trajectory tracking control of wheeled mobile robots. In contrast to [11] which employs full-state feedback, Wu et al. [12] developed a finite-time output feedback tracking control method. It has been demonstrated in numerous studies that finite-time control exhibits better robustness, faster convergence, and superior anti-interference capability compared to non-finite-time control methods [13], [14], [15]. However, there exists an undeniable fact that the convergence time is highly dependent on the initial system states. To address this, fixed-time control was proposed in [16], where the convergence time has an upper bound independent of the system's initial states. This control method has since been rapidly developed in the study of nonlinear systems [17], [18], [19], [20], [21]. For instance, Wang et al. [17] proposed a fuzzy fixed-time control scheme for higher-order nonlinear systems with sensor and actuator faults. The scheme was later extended to handle time-varying full-state constraints and actuator hysteresis [18]. Fixed-time control has

Received 30 July 2024; accepted 10 November 2024. Date of publication 21 November 2024; date of current version 29 November 2024. This article was recommended for publication by Associate Editor K. Tahara and Editor C. Gosselin upon evaluation of the reviewers' comments. This work was supported in part by the National Natural Science Foundation of China under Grant 52175046 and Grant 51939001 and in part by Sichuan Science and Technology Program under Grant 2024YFFK0037 and Grant 2024ZYD0165. *(Corresponding author: Qing Guo.)*

Chen Wang, Haoran Zhan, and Qing Guo are with the School of Aeronautics and Astronautics, University of Electronic Science and Technology of China, Chengdu 611731, China (e-mail: wangchen_chc@163.com; hr.zhan@std.uestc.edu.cn; guoqinguestc@uestc.edu.cn).

Tieshan Li is with the School of Automation Engineering, University of Electronic Science and Technology of China, Chengdu 611731, China (e-mail: litieshan073@uestc.edu.cn).

This letter has supplementary downloadable material available at https://doi.org/10.1109/LRA.2024.3504314, provided by the authors.

Digital Object Identifier 10.1109/LRA.2024.3504314

2377-3766 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Korea Advanced Inst of Science & Tech - KAIST. Downloaded on June 02,2025 at 07:35:32 UTC from IEEE Xplore. Restrictions apply.

also been applied to various practical systems, such as robotic arms [22], Euler-Lagrange systems [23], and quadrotor systems [24]. Despite these advances, there are still few fixed-time control schemes for wheeled mobile robots, which motivates this study.

Notice that the above control concerns for wheeled mobile robots are how to improve tracking accuracy precision and response speed under complex external conditions, and ignoring the control cost. In practical engineering applications, wheeled mobile robots have limited energy, so it is important to realize the balance between control performance and control cost [25]. To achieve this goal, a cost function can be specified and then the optimal control law can be found by solving the Hamilton-Jacobi-Bellman (HJB) equation. However, solving the HJB equation is difficult because it is a nonlinear partial differential equation. Fortunately, dynamic programming [26] provides a feasible way to find the optimal control solution. However, this approach suffers from the dimension disaster problem as the variable dimension increases. To solve this problem, adaptive dynamic programming (ADP) has been widely studied by scholars [27], [28], [29], [30], [31]. For example, Liu et al. [27] proposed an optimal robust guaranteed cost control method for uncertain nonlinear systems, using ADP to solve the optimal stabilization control problem. Later, Wang et al. [28] extended this method to address the optimal tracking control problem. In [30] and [31], the authors introduced ADP into backstepping control, proposing two different optimal control schemes for strict-feedback systems. Recently, to achieve optimal stability, some scholars have applied ADP to optimal control problems in real physical systems. For instance, Dong et al. [32] studied ADP-based optimal control for Euler-Lagrange systems. Xiao et al. [33] studied the attitude stabilization control problem of spacecraft using ADP and proposed an optimal fault-tolerant control method. Unfortunately, there are still relatively few studies on optimal trajectory tracking control for wheeled mobile robots, and even fewer that consider fixed-time stability. Motivated by the aforementioned studies, ADP-based fixed-time optimal trajectory tracking control for wheeled mobile robots is investigated, which has the following contributions.

- Many traditional ADP-based optimal control methods use the gradient descent algorithm to design neural network weight update laws, which require the persistent or finite excitation condition [32], [34], a requirement that is difficult to satisfy in practical applications. In this study, the adaptive control technique is employed to design the weight update law of the critic neural network, which not only eliminates the need for the excitation condition but also ensures fixed-time convergence of the weight estimation error.
- Most of the existing ADP-based optimal control methods can only achieve uniformly ultimately bounded stability [28], [29], [30], which may not satisfy the need for rapid stabilization in real-world applications. In this study, to ensure the fixed-time convergence of the trajectory tracking error in a wheeled mobile robot, a fixed-time optimal control strategy is proposed.

II. Motion Description of Wheeled Mobile Robot

As shown in Fig. 1, the position and attitude information of a differential drive mobile robot can be described by an earth-fixed frame $\{F_i\}$ and a body frame $\{F_b\}$, and the kinematic model of

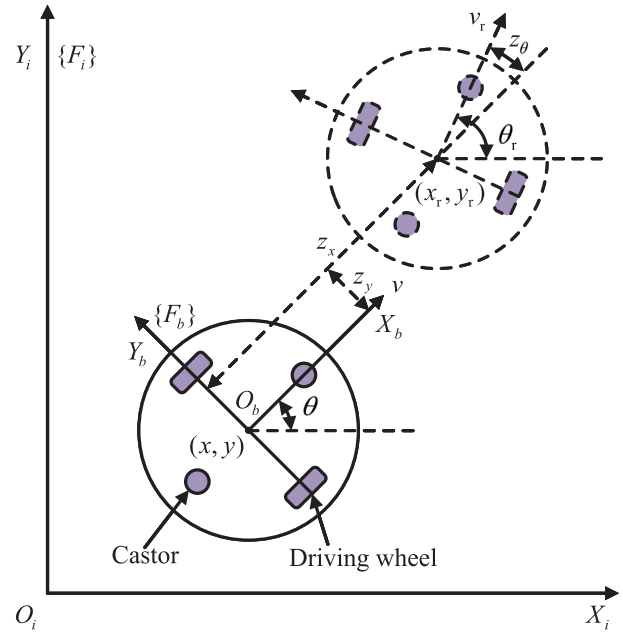


Fig. 1. Schematic diagram of wheeled mobile robot.

the robot is

$$\begin{cases} \dot{x} = v\cos\theta \\ \dot{y} = v\sin\theta \\ \dot{\theta} = \omega \end{cases} \tag{1}$$

where v denotes the linear velocity of the center of mass of the robot, and ω denotes the angular velocity; (x, y) and $(\dot{x}, \dot{y})$ represent the position and velocity in the earth-fixed frame, respectively; θ represents the heading direction; $\boldsymbol{u} = [v, \omega]^T \in \mathbb{R}^{2\times 1}$ is the control input of the robotic system which will be designed later. Furthermore, to characterize the trajectory tracking control problem, the following user-defined reference trajectory is introduced:

$$\begin{cases} \dot{x}_{\mathrm{r}} = v_{\mathrm{r}}\cos\theta_{\mathrm{r}} \\ \dot{y}_{\mathrm{r}} = v_{\mathrm{r}}\sin\theta_{\mathrm{r}} \\ \dot{\theta}_{\mathrm{r}} = \omega_{\mathrm{r}} \end{cases} \tag{2}$$

where x_{r}, y_{r} and θ_r are the desired system states; v_{r} and ω_{r} are the desired linear velocity and angular velocity, respectively.

Therefore, one can get a description of the trajectory tracking error of the robot as follows [35]:

$$\boldsymbol{z} = \begin{bmatrix} z_x \\ z_y \\ z_\theta \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_{\mathrm{r}} - x \\ y_{\mathrm{r}} - y \\ \theta_{\mathrm{r}} - \theta \end{bmatrix} \tag{3}$$

where $\boldsymbol{z} = [z_x, z_y, z_\theta]^{\mathrm{T}} \in \mathbb{R}^{3\times 1}$ represents the tracking error vector. Then, one can get the error dynamics of (3) as follows:

$$\begin{cases} \dot{z}_x = v_{\mathrm{r}}\cos z_\theta - v + z_y\omega \\ \dot{z}_y = v_{\mathrm{r}}\sin z_\theta - z_x\omega \\ \dot{z}_\theta = \omega_{\mathrm{r}} - \omega \end{cases}. \tag{4}$$

The control input is designed as $\boldsymbol{u} = \boldsymbol{u}_{\mathrm{f}} + \boldsymbol{u}_{\mathrm{o}}$, where $\boldsymbol{u}_{\mathrm{f}} = [v_{\mathrm{r}}\cos z_\theta, \omega_{\mathrm{r}}]^{\mathrm{T}} \in \mathbb{R}^{2\times 1}$ is the feedforward control input, and $\boldsymbol{u}_{\mathrm{o}} = [v_{\mathrm{o}}, \omega_{\mathrm{o}}]^{\mathrm{T}} \in \mathbb{R}^{2\times 1}$ is the feedback control input to be designed. Hence, the trajectory tracking control problem is transformed into an optimal regulation problem, and the error system

Authorized licensed use limited to: Korea Advanced Inst of Science & Tech - KAIST. Downloaded on June 02,2025 at 07:35:32 UTC from IEEE Xplore. Restrictions apply.

(4) is rewritten as

$$\dot{\boldsymbol{z}} = \boldsymbol{\mathcal{A}}(\boldsymbol{z}) + \boldsymbol{\mathcal{B}}(\boldsymbol{z})\boldsymbol{u}_{\mathrm{o}} \tag{5a}$$

$$\boldsymbol{\mathcal{A}}(\boldsymbol{z}) = [\omega_{\mathrm{r}} z_y \quad v_{\mathrm{r}} \sin z_\theta - \omega_{\mathrm{r}} z_x \quad 0]^{\mathrm{T}} \tag{5b}$$

$$\boldsymbol{\mathcal{B}}(\boldsymbol{z}) = \begin{bmatrix} -1 & 0 & 0 \\ z_y & -z_x & -1 \end{bmatrix}^{\mathrm{T}} \tag{5c}$$

where $\boldsymbol{\mathcal{A}}(\boldsymbol{z})$ is locally Lipchitz with respect to $\boldsymbol{z}$, and it is assumed that there exists a positive constant ψ_B such that $\|\boldsymbol{\mathcal{B}}(\boldsymbol{z})\| < \psi_B$ [28], [33]. For convenience, $\boldsymbol{\mathcal{A}}(\boldsymbol{z})$ and $\boldsymbol{\mathcal{B}}(\boldsymbol{z})$ will be abbreviated to $\boldsymbol{\mathcal{A}}$ and $\boldsymbol{\mathcal{B}}$, respectively.

III. ADP-Based Fixed-Time Optimal Control Design

The design of controller and system stability analysis requires the support of some mathematical lemmas, which are given as follows.

Lemma 1 [20]: For $\forall \varrho \in \mathbb{R}$, $\forall \bar{\varrho} \in \mathbb{R}$, $o > 0$, $\mu > 0$, and $1 - \mu > 0$, one has

$$|\bar{\varrho}|^{\mu}|\varrho|^{1-\mu} \leq (1-\mu)o^{\frac{-\mu}{1-\mu}}|\varrho| + \mu o|\bar{\varrho}|. \tag{6}$$

Lemma 2 [20]: For $b_i > 0$, $i = 1, 2, \ldots, n$, $0 < a_1 < 1$, and $a_2 \geq 1$, one has

$$\left(\sum_{i=1}^{n} b_i\right)^{a_1} \leq \sum_{i=1}^{n} b_i^{a_1}, \quad n^{1-a_2}\left(\sum_{i=1}^{n} b_i\right)^{a_2} \leq \sum_{i=1}^{n} b_i^{a_2}. \tag{7}$$

Lemma 3 [36]: For $\forall a_1 \in \mathbb{R}$ and $a_2 > 0$, one has

$$0 \leq |a_1| - a_1 \tanh\left(\frac{a_1}{a_2}\right) \leq 0.2785 a_2. \tag{8}$$

A. Ideal Optimal Controller

The control objective is to design the optimal control law $\boldsymbol{u}_{\mathrm{o}}$ that effectively minimizes the prescribed cost function

$$\mathcal{J}(\boldsymbol{z}) = \int_t^{\infty} \left[\boldsymbol{u}_{\mathrm{o}}^{\mathrm{T}}(\boldsymbol{z})\boldsymbol{\mathcal{R}}\boldsymbol{u}_{\mathrm{o}}(\boldsymbol{z}) + \boldsymbol{z}^{\mathrm{T}}(\tau)\boldsymbol{\mathcal{Q}}\boldsymbol{z}(\tau)\right] \mathrm{d}\tau \tag{9}$$

where $\boldsymbol{\mathcal{R}} \in \mathbb{R}^{2\times 2}$ and $\boldsymbol{\mathcal{Q}} \in \mathbb{R}^{3\times 3}$ are positive definite matrices.

Assuming that the optimal control law is $\boldsymbol{u}_{\mathrm{o}}^{\star}$ which is $\mathbb{C}^1$, one can obtain the optimal cost function

$$\mathcal{J}^{\star}(\boldsymbol{z}) = \int_t^{\infty} \left[\boldsymbol{u}_{\mathrm{o}}^{\star\mathrm{T}}(\boldsymbol{z})\boldsymbol{\mathcal{R}}\boldsymbol{u}_{\mathrm{o}}^{\star}(\boldsymbol{z}) + \boldsymbol{z}^{\mathrm{T}}(\tau)\boldsymbol{\mathcal{Q}}\boldsymbol{z}(\tau)\right] \mathrm{d}\tau. \tag{10}$$

Then, the HJB equation is obtained as follows:

$$\begin{aligned}\mathcal{H}\left(\boldsymbol{z}, \boldsymbol{u}_{\mathrm{o}}^{\star}, \nabla\mathcal{J}^{\star}(\boldsymbol{z})\right) = \nabla\mathcal{J}^{\star\mathrm{T}}(\boldsymbol{z})\left(\boldsymbol{\mathcal{A}} + \boldsymbol{\mathcal{B}}\boldsymbol{u}_{\mathrm{o}}^{\star}\right) + \boldsymbol{z}^{\mathrm{T}}\boldsymbol{\mathcal{Q}}\boldsymbol{z} \\ + \boldsymbol{u}_{\mathrm{o}}^{\star\mathrm{T}}\boldsymbol{\mathcal{R}}\boldsymbol{u}_{\mathrm{o}}^{\star} = 0\end{aligned} \tag{11}$$

where $\nabla\mathcal{J}^{\star\mathrm{T}}(\boldsymbol{z}) = [\partial\mathcal{J}^{\star}(\boldsymbol{z})/\partial\boldsymbol{z}]^{\mathrm{T}} \in \mathbb{R}^{1\times 3}$, $\partial\mathcal{J}^{\star}(\boldsymbol{z})/\partial\boldsymbol{z} = [\partial\mathcal{J}^{\star}(\boldsymbol{z})/\partial z_x, \partial\mathcal{J}^{\star}(\boldsymbol{z})/\partial z_y, \partial\mathcal{J}^{\star}(\boldsymbol{z})/\partial z_\theta]^{\mathrm{T}} \in \mathbb{R}^{3\times 1}$.

According to $\partial\mathcal{H}(\boldsymbol{z}, \boldsymbol{u}_{\mathrm{o}}^{\star}, \nabla\mathcal{J}^{\star}(\boldsymbol{z}))/\partial\boldsymbol{u}_{\mathrm{o}}^{\star} = \boldsymbol{0}_{2\times 1}$, the ideal optimal control law is derived as

$$\boldsymbol{u}_{\mathrm{o}}^{\star} = -\frac{1}{2}\boldsymbol{\mathcal{R}}^{-1}\boldsymbol{\mathcal{B}}^{\mathrm{T}}\nabla\mathcal{J}^{\star}(\boldsymbol{z}). \tag{12}$$

Finally, applying (12) to (11) yields

$$\begin{aligned}\mathcal{H}\left(\boldsymbol{z}, \boldsymbol{u}_{\mathrm{o}}^{\star}, \nabla\mathcal{J}^{\star}(\boldsymbol{z})\right) = -\frac{1}{4}\nabla\mathcal{J}^{\star\mathrm{T}}(\boldsymbol{z})\boldsymbol{\mathcal{B}}\boldsymbol{\mathcal{R}}^{-1}\boldsymbol{\mathcal{B}}^{\mathrm{T}}\nabla\mathcal{J}^{\star}(\boldsymbol{z}) \\ + \nabla\mathcal{J}^{\star\mathrm{T}}(\boldsymbol{z})\boldsymbol{\mathcal{A}} + \boldsymbol{z}^{\mathrm{T}}\boldsymbol{\mathcal{Q}}\boldsymbol{z}.\end{aligned} \tag{13}$$

Remark 1: The analytical solution of the optimal control law (12) requires solving the partial derivative function $\nabla\mathcal{J}^{\star}(\boldsymbol{z})$ through the HJB (13). Unfortunately, solving the (13) directly can be quite challenging, thereby rendering it impractical to directly utilize the optimal control law (12) to achieve optimal performance for the error system (5).

B. ADP-Based Fixed-Time Optimal Tracking Controller Design

To overcome the difficulty of solving the HJB equation, a critic neural network is used to estimate the optimal cost function $\mathcal{J}^{\star}(\boldsymbol{z})$ defined on a compact set $\boldsymbol{z} \in \boldsymbol{\Theta} \in \mathbb{R}^{3\times 1}$. It can be described as

$$\mathcal{J}^{\star}(\boldsymbol{z}) = \boldsymbol{\varphi}^{\mathrm{T}}(\boldsymbol{z})\boldsymbol{\mathcal{W}}^{\star} + \sigma(\boldsymbol{z}) \tag{14}$$

where $\boldsymbol{\mathcal{W}}^{\star} = [\mathcal{W}_1^{\star}, \mathcal{W}_2^{\star}, \ldots, \mathcal{W}_l^{\star}]^{\mathrm{T}} \in \mathbb{R}^{l\times 1}$ is the ideal weight vector; $\sigma(\boldsymbol{z}) \in \mathbb{R}$ is the estimation error; $\boldsymbol{\varphi}(\boldsymbol{z}) = [\varphi_1(\boldsymbol{z}), \varphi_2(\boldsymbol{z}), \ldots, \varphi_l(\boldsymbol{z})]^{\mathrm{T}} \in \mathbb{R}^{l\times 1}$ is the basis function vector.

Hence, the optimal control law (12) can be reformulated as

$$\boldsymbol{u}_{\mathrm{o}}^{\star} = -\frac{1}{2}\boldsymbol{\mathcal{R}}^{-1}\boldsymbol{\mathcal{B}}^{\mathrm{T}}\left[\nabla\boldsymbol{\varphi}^{\mathrm{T}}(\boldsymbol{z})\boldsymbol{\mathcal{W}}^{\star} + \nabla\sigma(\boldsymbol{z})\right] \tag{15}$$

where $\nabla\boldsymbol{\varphi}^{\mathrm{T}}(\boldsymbol{z}) = [\partial\boldsymbol{\varphi}(\boldsymbol{z})/\partial\boldsymbol{z}]^{\mathrm{T}} \in \mathbb{R}^{3\times l}$, $\partial\boldsymbol{\varphi}(\boldsymbol{z})/\partial\boldsymbol{z} = [\partial\varphi_1(\boldsymbol{z})/\partial\boldsymbol{z}, \partial\varphi_2(\boldsymbol{z})/\partial\boldsymbol{z}, \ldots, \partial\varphi_l(\boldsymbol{z})/\partial\boldsymbol{z}]^{\mathrm{T}} \in \mathbb{R}^{l\times 3}$, $\partial\varphi_i(\boldsymbol{z})/\partial\boldsymbol{z} = [\partial\varphi_i(\boldsymbol{z})/\partial z_x, \varphi_i(\boldsymbol{z})/\partial z_y, \varphi_i(\boldsymbol{z})/\partial z_\theta]^{\mathrm{T}} \in \mathbb{R}^{3\times 1}$, $i = 1, 2, \ldots, l$, and $\nabla\sigma(\boldsymbol{z}) = \partial\sigma(\boldsymbol{z})/\partial\boldsymbol{z} \in \mathbb{R}^{3\times 1}$.

Since $\boldsymbol{\mathcal{W}}^{\star}$ is unknown, the corresponding estimation $\hat{\boldsymbol{\mathcal{W}}} = [\hat{\mathcal{W}}_1, \hat{\mathcal{W}}_2, \ldots, \hat{\mathcal{W}}_l]^{\mathrm{T}} \in \mathbb{R}^{l\times 1}$ is introduced. Thus, the estimation of $\boldsymbol{u}_{\mathrm{o}}^{\star}$ is

$$\hat{\boldsymbol{u}}_{\mathrm{o}}^{\star} = -\frac{1}{2}\boldsymbol{\mathcal{R}}^{-1}\boldsymbol{\mathcal{B}}^{\mathrm{T}}\nabla\boldsymbol{\varphi}^{\mathrm{T}}(\boldsymbol{z})\hat{\boldsymbol{\mathcal{W}}}. \tag{16}$$

Meanwhile, $\hat{\boldsymbol{\mathcal{W}}}$ is updated by the adaptive update law

$$\dot{\hat{\boldsymbol{\mathcal{W}}}} = \frac{1}{2}\boldsymbol{\Gamma}\left[\nabla\boldsymbol{\varphi}(\boldsymbol{z})\boldsymbol{\mathcal{B}}\boldsymbol{\mathcal{R}}^{-1}\boldsymbol{\mathcal{B}}^{\mathrm{T}}\boldsymbol{z} - \kappa_1\hat{\boldsymbol{\mathcal{W}}} - \kappa_2\left(\hat{\boldsymbol{\mathcal{W}}}^{\mathrm{T}}\hat{\boldsymbol{\mathcal{W}}}\right)\hat{\boldsymbol{\mathcal{W}}}\right] \tag{17}$$

where $\boldsymbol{\Gamma} \in \mathbb{R}^{l\times l}$ is a positive definite matrix, and κ_1 and κ_2 are positive constants.

Based on the above design, the ADP-based fixed-time optimal tracking controller is constructed as follows:

$$\boldsymbol{u}_{\mathrm{o}} = \hat{\boldsymbol{u}}_{\mathrm{o}}^{\star} - \boldsymbol{\mathcal{B}}^{\dagger}\left[\boldsymbol{\lambda}\tanh\left(\frac{\boldsymbol{z}}{\rho}\right) + \boldsymbol{\mu}\boldsymbol{z} + \boldsymbol{\alpha}\boldsymbol{z}^{\frac{p}{q}} + \boldsymbol{\beta}\boldsymbol{z}^3\right] \tag{18}$$

where $\boldsymbol{\mathcal{B}}^{\dagger}$ the pseudo-inverse of $\boldsymbol{\mathcal{B}}$; $\boldsymbol{\lambda}$, $\boldsymbol{\mu}$, $\boldsymbol{\alpha}$, and $\boldsymbol{\beta}$ are positive definite matrices with appropriate dimensions; $\tanh(\boldsymbol{z}/\rho) = [\tanh(z_x/\rho), \tanh(z_y/\rho), \tanh(z_\theta/\rho)]^{\mathrm{T}}$, $\boldsymbol{z}^3 = [z_x^3, z_y^3, z_\theta^3]^{\mathrm{T}}$, $\boldsymbol{z}^{p/q} = [z_x^{p/q}, z_y^{p/q}, z_\theta^{p/q}]^{\mathrm{T}}$; p and q are positive odd integers satisfying $p < q$.

Up to now, the fixed-time optimal controller has been designed. To gain a deeper understanding of the controller's structure, its primary framework is presented in Fig. 2.

C. System Stability Analysis

Theorem 1: Consider the tracking error system described by (5a)–(5c) for the nonholonomic mobile robot. The proposed ADP-based fixed-time optimal tracking controller (18) can stabilize the origin of the error system in fixed time independent of the initial condition. Specifically, the tracking error and the

Authorized licensed use limited to: Korea Advanced Inst of Science & Tech - KAIST. Downloaded on June 02,2025 at 07:35:32 UTC from IEEE Xplore. Restrictions apply.

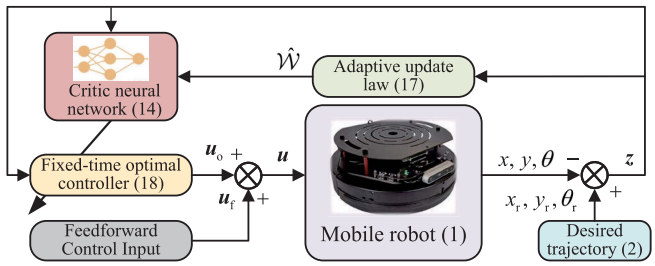


Fig. 2. Controller structure diagram.

weight estimation error of the neural network can converge to a neighborhood of zero in fixed time $T_{\max}$.

Proof: A Lyapunov function is chosen as

$$V(\boldsymbol{z}, \tilde{\boldsymbol{\mathcal{W}}}) = \frac{1}{2}\boldsymbol{z}^{\mathrm{T}}\boldsymbol{z} + \frac{1}{2}\tilde{\boldsymbol{\mathcal{W}}}^{\mathrm{T}}\boldsymbol{\Gamma}^{-1}\tilde{\boldsymbol{\mathcal{W}}} \tag{19}$$

where $\tilde{\boldsymbol{\mathcal{W}}} = \boldsymbol{\mathcal{W}}^{\star} - \hat{\boldsymbol{\mathcal{W}}} = [\tilde{\mathcal{W}}_1, \tilde{\mathcal{W}}_2, \ldots, \tilde{\mathcal{W}}_l]^{\mathrm{T}} \in \mathbb{R}^{l\times 1}$ is the weight estimation error; $\boldsymbol{\Gamma}^{-1}$ is the inverse of $\boldsymbol{\Gamma}$.

Calculating the time derivative of (19) yields

$$\begin{aligned}\dot{V} &= \boldsymbol{z}^{\mathrm{T}}\Big\{\boldsymbol{\mathcal{B}}\Big[\frac{1}{2}\boldsymbol{\mathcal{R}}^{-1}\boldsymbol{\mathcal{B}}^{\mathrm{T}}\nabla\boldsymbol{\varphi}^{\mathrm{T}}(\boldsymbol{z})\left(\boldsymbol{\mathcal{W}}^{\star} - \hat{\boldsymbol{\mathcal{W}}}\right) + \boldsymbol{u}_{\mathrm{o}}^{\star} \\ &\quad + \frac{1}{2}\boldsymbol{\mathcal{R}}^{-1}\boldsymbol{\mathcal{B}}^{\mathrm{T}}\nabla\sigma(\boldsymbol{z})\Big] + \boldsymbol{\mathcal{A}} - \boldsymbol{\mu}\boldsymbol{z} - \boldsymbol{\alpha}\boldsymbol{z}^{\frac{p}{q}} - \boldsymbol{\beta}\boldsymbol{z}^3 \\ &\quad - \boldsymbol{\lambda}\tanh\left(\frac{\boldsymbol{z}}{\rho}\right)\Big\} - \tilde{\boldsymbol{\mathcal{W}}}^{\mathrm{T}}\boldsymbol{\Gamma}^{-1}\dot{\hat{\boldsymbol{\mathcal{W}}}} \\ &= \tilde{\boldsymbol{\mathcal{W}}}^{\mathrm{T}}\left[\frac{1}{2}\nabla\boldsymbol{\varphi}(\boldsymbol{z})\boldsymbol{\mathcal{B}}\boldsymbol{\mathcal{R}}^{-1}\boldsymbol{\mathcal{B}}^{\mathrm{T}}\boldsymbol{z} - \boldsymbol{\Gamma}^{-1}\dot{\hat{\boldsymbol{\mathcal{W}}}}\right] \\ &\quad + \boldsymbol{z}^{\mathrm{T}}\Big[\boldsymbol{\mathcal{B}}\boldsymbol{u}_{\mathrm{o}}^{\star} + \boldsymbol{\mathcal{A}} + \frac{1}{2}\boldsymbol{\mathcal{B}}\boldsymbol{\mathcal{R}}^{-1}\boldsymbol{\mathcal{B}}^{\mathrm{T}}\nabla\sigma(\boldsymbol{z}) \\ &\quad - \boldsymbol{\mu}\boldsymbol{z} - \boldsymbol{\lambda}\tanh\left(\frac{\boldsymbol{z}}{\rho}\right) - \boldsymbol{\alpha}\boldsymbol{z}^{\frac{p}{q}} - \boldsymbol{\beta}\boldsymbol{z}^3\Big].\end{aligned} \tag{20}$$

Then, substituting (17) into (20) yields

$$\begin{aligned}\dot{V} &= \kappa_1\tilde{\boldsymbol{\mathcal{W}}}^{\mathrm{T}}\hat{\boldsymbol{\mathcal{W}}} + \kappa_2\left(\tilde{\boldsymbol{\mathcal{W}}}^{\mathrm{T}}\hat{\boldsymbol{\mathcal{W}}}\right)\hat{\boldsymbol{\mathcal{W}}}^{\mathrm{T}}\hat{\boldsymbol{\mathcal{W}}} \\ &\quad + \boldsymbol{z}^{\mathrm{T}}\Big[\boldsymbol{\mathcal{B}}\boldsymbol{u}_{\mathrm{o}}^{\star} + \boldsymbol{\mathcal{A}} + \frac{1}{2}\boldsymbol{\mathcal{B}}\boldsymbol{\mathcal{R}}^{-1}\boldsymbol{\mathcal{B}}^{\mathrm{T}}\nabla\sigma(\boldsymbol{z}) \\ &\quad - \boldsymbol{\mu}\boldsymbol{z} - \boldsymbol{\lambda}\tanh\left(\frac{\boldsymbol{z}}{\rho}\right) - \boldsymbol{\alpha}\boldsymbol{z}^{\frac{p}{q}} - \boldsymbol{\beta}\boldsymbol{z}^3\Big].\end{aligned} \tag{21}$$

Based on the inequalities (39) and (42) obtained in appendix, we can get

$$\begin{aligned}\dot{V} \le &-\delta_1\left(\frac{1}{2}\tilde{\boldsymbol{\mathcal{W}}}^{\mathrm{T}}\boldsymbol{\Gamma}^{-1}\tilde{\boldsymbol{\mathcal{W}}}\right)^{\frac{p+q}{2q}} - \phi_1\left(\frac{1}{2}\tilde{\boldsymbol{\mathcal{W}}}^{\mathrm{T}}\boldsymbol{\Gamma}^{-1}\tilde{\boldsymbol{\mathcal{W}}}\right)^2 \\ &+ \boldsymbol{z}^{\mathrm{T}}\Big[\boldsymbol{\mathcal{B}}\boldsymbol{u}_{\mathrm{o}}^{\star} + \frac{1}{2}\boldsymbol{\mathcal{B}}\boldsymbol{\mathcal{R}}^{-1}\boldsymbol{\mathcal{B}}^{\mathrm{T}}\nabla\sigma(\boldsymbol{z}) + \boldsymbol{\mathcal{A}} - \boldsymbol{\mu}\boldsymbol{z} \\ &-\boldsymbol{\lambda}\tanh\left(\frac{\boldsymbol{z}}{\rho}\right) - \boldsymbol{\alpha}\boldsymbol{z}^{\frac{p}{q}} - \boldsymbol{\beta}\boldsymbol{z}^3\Big] + \varsigma\end{aligned} \tag{22}$$

with

$$\begin{aligned}\varsigma = &\frac{1}{2}\kappa_1\boldsymbol{\mathcal{W}}^{\star\mathrm{T}}\boldsymbol{\mathcal{W}}^{\star} + \frac{q-p}{2q}\left(\frac{p+q}{2q}\right)^{\frac{p+q}{q-p}} \\ &+ \sum_{i=1}^{l}\left(\frac{\kappa_2\mathcal{W}_i^{\star 4}}{12} + \frac{3\kappa_2\mathcal{W}_i^{\star 4}}{4\varpi_i^4}\right).\end{aligned} \tag{23}$$

Since $\mathcal{A}(\boldsymbol{z})$ is locally Lipchitz, there exists ψ_A such that $\|\mathcal{A}\| < \psi_A\|\boldsymbol{z}\|$ holds, where $\|\cdot\|$ stands for the Euclidean norm. Therefore, if a suitable $\boldsymbol{\mu}$ is chosen to satisfy the condition $\vartheta_{\min}(\boldsymbol{\mu}) \ge \psi_A$, the inequality $\boldsymbol{z}^{\mathrm{T}}(\boldsymbol{\mathcal{A}} - \boldsymbol{\mu}\boldsymbol{z}) \le \psi_A\|\boldsymbol{z}\|^2 - \vartheta_{\min}(\boldsymbol{\mu})\|\boldsymbol{z}\|^2 \le 0$ holds, where $\vartheta_{\min}(\boldsymbol{\mu})$ represents the minimum eigenvalue of $\boldsymbol{\mu}$. By *Lemma 3*, it can be obtained that $-\boldsymbol{z}^{\mathrm{T}}\boldsymbol{\lambda}\tanh(\boldsymbol{z}/\rho) \le -|\boldsymbol{z}|^{\mathrm{T}}\boldsymbol{\lambda}\boldsymbol{\iota} + 0.2785\rho\boldsymbol{\iota}^{\mathrm{T}}\boldsymbol{\lambda}\boldsymbol{\iota}$ with $\boldsymbol{\iota} = [1, 1, 1]^{\mathrm{T}}$ and $|\boldsymbol{z}| = [|z_x|, |z_y|, |z_\theta|]^{\mathrm{T}}$. Theoretically, the optimal control law $\boldsymbol{u}_{\mathrm{o}}^{\star}$ exists and can be approximated using critic neural networks. Therefore, it is reasonable to assume that $\boldsymbol{u}_{\mathrm{o}}^{\star}$ and $\nabla\sigma(\boldsymbol{z})$ are bounded [33]. Further, it can be obtained that $\boldsymbol{z}^{\mathrm{T}}[\boldsymbol{\mathcal{B}}\boldsymbol{u}_{\mathrm{o}}^{\star} + \frac{1}{2}\boldsymbol{\mathcal{B}}\boldsymbol{\mathcal{R}}^{-1}\boldsymbol{\mathcal{B}}^{\mathrm{T}}\nabla\sigma(\boldsymbol{z})] \le |\boldsymbol{z}|^{\mathrm{T}}\boldsymbol{\tau}$ since $\boldsymbol{\mathcal{B}}$ is bounded, where $\boldsymbol{\tau} = [\tau_1, \tau_2, \tau_3]^{\mathrm{T}}$, $\tau_i > 0$, and $i = 1, 2, 3$. Thus, a suitable $\boldsymbol{\lambda}$ can make $|\boldsymbol{z}|^{\mathrm{T}}\boldsymbol{\tau} - |\boldsymbol{z}|^{\mathrm{T}}\boldsymbol{\lambda}\boldsymbol{\iota} \le 0$ hold. Based on the above discussion, if suitable parameters $\boldsymbol{\mu}$ and $\boldsymbol{\lambda}$ are defined, the following inequality holds:

$$\begin{aligned}\boldsymbol{z}^{\mathrm{T}}\Big[&\boldsymbol{\mathcal{B}}\boldsymbol{u}_{\mathrm{o}}^{\star} + \frac{1}{2}\boldsymbol{\mathcal{B}}\boldsymbol{\mathcal{R}}^{-1}\boldsymbol{\mathcal{B}}^{\mathrm{T}}\nabla\sigma(\boldsymbol{z}) + \boldsymbol{\mathcal{A}} - \boldsymbol{\mu}\boldsymbol{z} \\ &-\boldsymbol{\lambda}\tanh\left(\frac{\boldsymbol{z}}{\rho}\right)\Big] \le 0.2785\rho\boldsymbol{\iota}^{\mathrm{T}}\boldsymbol{\lambda}\boldsymbol{\iota}.\end{aligned} \tag{24}$$

Then, substituting (24) into (22) yields

$$\begin{aligned}\dot{V} \le &-\delta_1\left(\frac{1}{2}\tilde{\boldsymbol{\mathcal{W}}}^{\mathrm{T}}\boldsymbol{\Gamma}^{-1}\tilde{\boldsymbol{\mathcal{W}}}\right)^{\frac{p+q}{2q}} - \phi_1\left(\frac{1}{2}\tilde{\boldsymbol{\mathcal{W}}}^{\mathrm{T}}\boldsymbol{\Gamma}^{-1}\tilde{\boldsymbol{\mathcal{W}}}\right)^2 \\ &-\delta_2\left(\frac{1}{2}\boldsymbol{z}^{\mathrm{T}}\boldsymbol{z}\right)^{\frac{p+q}{2q}} - \phi_2\left(\frac{1}{2}\boldsymbol{z}^{\mathrm{T}}\boldsymbol{z}\right)^2 + \bar{\varsigma} \\ \le &-\delta V^{\frac{p+q}{2q}} - \phi V^2 + \bar{\varsigma}\end{aligned} \tag{25}$$

where $\delta_2 = 2^{(p+q)/2q}\vartheta_{\min}(\boldsymbol{\alpha})$, $\phi_2 = 4\vartheta_{\min}(\boldsymbol{\beta})$, $\bar{\varsigma} = \varsigma + 0.2785\rho\boldsymbol{\iota}^{\mathrm{T}}\boldsymbol{\lambda}\boldsymbol{\iota}$, $\delta = \min\{\delta_1, \delta_2\}$, $\phi = 2\min\{\phi_1, \phi_2\}$, and $\vartheta_{\min}(\boldsymbol{\alpha})$ and $\vartheta_{\min}(\boldsymbol{\beta})$ are the minimum eigenvalues of $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$, respectively.

According to the *Lemma 2.2* in [37], the Lyapunov function V can converge to the region described as

$$V \le \min\left\{\left[\frac{\bar{\varsigma}}{(1-\varepsilon)\delta}\right]^{\frac{2q}{p+q}}, \left[\frac{\bar{\varsigma}}{(1-\varepsilon)\phi}\right]^{\frac{1}{2}}\right\} \tag{26}$$

and the convergence time $T(\boldsymbol{z}(0), \tilde{\boldsymbol{\mathcal{W}}}(0))$ satisfies

$$T(\boldsymbol{z}(0), \tilde{\boldsymbol{\mathcal{W}}}(0)) \le T_{\max} := \frac{2q}{\delta\varepsilon\,(q-p)} + \frac{1}{\phi\varepsilon} \tag{27}$$

where $\varepsilon \in (0, 1)$, $\boldsymbol{z}(0)$ and $\tilde{\boldsymbol{\mathcal{W}}}(0)$ are the initial errors.

According to (19) and (27), as $t \to T_{\max}$, the robot's tracking error and neural network weight estimation error satisfy

$$\|\boldsymbol{z}\| \le \min\left\{\sqrt{2}\left[\frac{\bar{\varsigma}}{(1-\varepsilon)\delta}\right]^{\frac{q}{p+q}}, \left[\frac{4\bar{\varsigma}}{(1-\varepsilon)\phi}\right]^{\frac{1}{4}}\right\} \tag{28}$$

Authorized licensed use limited to: Korea Advanced Inst of Science & Tech - KAIST. Downloaded on June 02,2025 at 07:35:32 UTC from IEEE Xplore. Restrictions apply.

$$\|\tilde{\boldsymbol{\mathcal{W}}}\| \leq \frac{\min\left\{\sqrt{2}\left[\frac{\bar{\varsigma}}{(1-\varepsilon)\delta}\right]^{\frac{q}{p+q}}, \left\{\frac{4\bar{\varsigma}}{(1-\varepsilon)\phi}\right\}^{\frac{1}{4}}\right\}}{\vartheta_{\min}(\boldsymbol{\Gamma}^{-1})} \tag{29}$$

where $\vartheta_{\min}(\boldsymbol{\Gamma}^{-1})$ represents the maximum eigenvalue of $\boldsymbol{\Gamma}^{-1}$.

Then, the error between the ideal optimal controller (15) and the ADP-based fixed-time optimal controller (18) needs to be analyzed. According to (16) and (18), one has

$$\begin{aligned}\boldsymbol{u}_{\mathrm{o}} - \boldsymbol{u}_{\mathrm{o}}^{\star} = &\frac{1}{2}\boldsymbol{\mathcal{R}}^{-1}\boldsymbol{\mathcal{B}}^{\mathrm{T}}\left[\nabla\boldsymbol{\varphi}^{\mathrm{T}}(\boldsymbol{z})\tilde{\boldsymbol{\mathcal{W}}} + \nabla\sigma(\boldsymbol{z})\right] \\ &- \boldsymbol{\mathcal{B}}^{\dagger}\left[\boldsymbol{\lambda}\tanh\left(\frac{\boldsymbol{z}}{\rho}\right) + \boldsymbol{\mu}\boldsymbol{z} + \boldsymbol{\alpha}\boldsymbol{z}^{\frac{p}{q}} + \boldsymbol{\beta}\boldsymbol{z}^{3}\right].\end{aligned} \tag{30}$$

Notice that $\boldsymbol{\mathcal{B}}$, $\boldsymbol{\mathcal{B}}^{\dagger}$, $\nabla\boldsymbol{\varphi}^{\mathrm{T}}(\boldsymbol{z})$, and $\nabla\sigma(\boldsymbol{z})$ are function vectors with respect to $\boldsymbol{z}$. Moreover, according to (28) and (29), $\boldsymbol{z}$ and $\tilde{\boldsymbol{\mathcal{W}}}$ are bounded. Hence, it follows that $\|\boldsymbol{u}_{\mathrm{o}} - \boldsymbol{u}_{\mathrm{o}}^{\star}\|$ is bounded, which implies that the fixed-time optimal controller $\boldsymbol{u}_{\mathrm{o}}$ designed using critic neural networks can fully approximate the ideal optimal controller $\boldsymbol{u}_{\mathrm{o}}^{\star}$ by adjusting the relevant customized parameters. The proof of *Theorem 1* is complete. ■

Remark 2: In some existing studies on ADP-based optimal control [32], [34], the neural network weight update law is designed based on the gradient descent algorithm, which requires the persistent/finite excitation condition. Although [38] removed the above excitation condition by introducing a positive definite function, it still requires neural networks with the same actor-critic structure as [34]. This complex structure is not favorable for engineering applications. In contrast, only a critic neural network is used in this study, which not only removes the excitation condition, but also simplifies the controller structure.

Remark 3: Although the optimal control performance can be obtained by some existing ADP-based optimal control methods [28], [29], [30], the convergence speed of the system has not been concerned. Specifically, they can only guarantee uniformly ultimately bounded stability, so the system takes take an infinitely long time to enter the final stable state. In this study, we concurrently consider both the convergence speed and the optimal control performance of the system, ensuring that both aspects are optimized for improved overall performance.

Remark 4: Utilizing the Lyapunov stability theory, it has been shown that both the tracking error and the weight estimation error of the system can converge to the compact sets described by (28) and (29), respectively. According to (27), the convergence time of the system is bounded, and the boundary is independent of the system's initial state. This is in contrast to the uniformly ultimately bounded stability and the finite-time stability [30], [31], [32]. From (27)–(29), one can choose the relevant parameters to achieve larger values of δ and ϕ and smaller values of $\bar{\varsigma}$, thereby reducing the tracking and estimation errors. On the other hand, convergence can be accelerated by increasing the values of δ and ϕ. For instance, δ and ϕ can be increased by raising $\vartheta_{\min}(\boldsymbol{\alpha})$, $\vartheta_{\min}(\boldsymbol{\beta})$, κ_1, and κ_2, and vice versa. However, excessively large values of δ and ϕ may result in a large control input, potentially causing system chatter or even instability. Therefore, all parameters must be selected carefully to optimize the control performance.

IV. Simulations and Experiments

In order to verify the feasibility of the proposed optimal controller, simulations and physical experiments are conducted. We compare the proposed controller with two other controllers. One of them is the classical backstepping controller (marked as BC) which is designed as follows [35]:

$$\begin{bmatrix} v \\ \omega \end{bmatrix} = \begin{bmatrix} \xi_1 z_x + v_{\mathrm{r}}\cos z_\theta \\ \omega_{\mathrm{r}} + \xi_2 v_{\mathrm{r}} z_y + \xi_3 v_{\mathrm{r}}\sin z_\theta \end{bmatrix}^{\mathrm{T}} \tag{31}$$

where ξ_1, ξ_2, and ξ_3 are positive constants; z_θ, z_x, and z_y are given in (3). The other one is the ADP-based optimal controller (marked as ADP-UUB) with uniformly ultimately bounded stability, which is obtained by modifying the proposed controller. To be specific, replace (17) and (18) with

$$\dot{\hat{\boldsymbol{\mathcal{W}}}} = \frac{1}{2}\boldsymbol{\Gamma}\left[\nabla\boldsymbol{\varphi}(\boldsymbol{z})\boldsymbol{\mathcal{B}}\boldsymbol{\mathcal{R}}^{-1}\boldsymbol{\mathcal{B}}^{\mathrm{T}}\boldsymbol{z} - \kappa_1\hat{\boldsymbol{\mathcal{W}}}\right] \tag{32}$$

and

$$\boldsymbol{u}_{\mathrm{o}} = \hat{\boldsymbol{u}}_{\mathrm{o}}^{\star} - \boldsymbol{\mathcal{B}}^{\dagger}\left[\boldsymbol{\lambda}\tanh\left(\frac{\boldsymbol{z}}{\rho}\right) + \boldsymbol{\mu}\boldsymbol{z} + \boldsymbol{\alpha}\boldsymbol{z}\right] \tag{33}$$

where the relevant parameter definitions are the same as (17) and (18). In addition, all the remaining designs are consistent with the proposed controller. To ensure the fairness of the comparison, the control parameters of ADP-UUB and the proposed controller are chosen consistently in the follow-up simulations and experiments.

Remark 5: Similar to the proof of *Theorem 1*, the ADP-UUB allows the Lyapunov function (19) to satisfy the inequality $\dot{V} \leq -\delta_1 V + \delta_2$ with $\delta_1 > 0$ and $\delta_2 > 0$, which implies that the ADP-UUB could make the error system (4) uniformly ultimately bounded. Moreover, since the ADP-UUB is designed in the framework of optimal control, the system can also obtain optimal control performance. It is worth mentioning that ADP-UUB allows the robot's tracking error to converge exponentially [30]. Theoretically, as time tends to infinity, the tracking error converges to a neighborhood of the origin. In contrast, the control method proposed in this study can achieve tracking error convergence in fixed time.

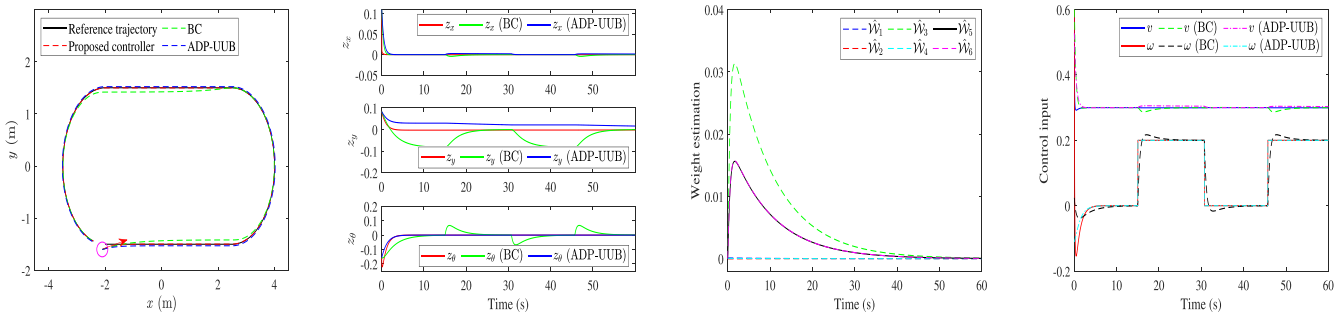
A. Simulation Results

The control task is to drive the mobile robot to move following the reference trajectory defined by $[x_{\mathrm{r}}(0), y_{\mathrm{r}}(0), \theta_{\mathrm{r}}(0)]^{\mathrm{T}} = [-2, -1.5, 0]^{\mathrm{T}}$, $v_{\mathrm{r}} = 0.3$ m/s, and ω_{r} is given as

$$\omega_{\mathrm{r}} = \begin{cases} 0 \text{ rad/s} & 0 \text{ s} \leq t \leq 15 \text{ s} \\ 0.2 \text{ rad/s} & 15 \text{ s} < t \leq 5\pi + 15 \text{ s} \\ 0 \text{ rad/s} & 5\pi + 15 < t \leq 5\pi + 30 \text{ s} \\ 0.2 \text{ rad/s} & 5\pi + 30 \text{ s} < t \leq 60 \text{ s} \end{cases}. \tag{34}$$

The initial position of the mobile robot is specified as $[x(0), y(0), \theta(0)]^{\mathrm{T}} = [-2.1, -1.6, \pi/20]^{\mathrm{T}}$. The basis function vector is designed as $\boldsymbol{\varphi}(\boldsymbol{z}) = [z_x^2, z_y^2, z_\theta^2, z_x z_y, z_y z_\theta, z_x z_\theta]^{\mathrm{T}}$. Hence, the corresponding partial derivative term is $\nabla\boldsymbol{\varphi}^{\mathrm{T}}(\boldsymbol{z}) = [2z_x, 0, 0, z_y, 0, z_\theta; 0, 2z_y, 0, z_x, z_\theta, 0; 0, 0, 2z_\theta, 0, z_y, z_x]$. The initial value of the weight estimation vector is $\hat{\boldsymbol{\mathcal{W}}}(0) = [0, 0, 0, 0, 0, 0]^{\mathrm{T}}$. The parameters of the proposed optimal controller are chosen as follows: $p = 17$, $q = 19$, $\boldsymbol{\mathcal{R}} = 2\boldsymbol{I}_3$, $\boldsymbol{\mathcal{Q}} = 10\boldsymbol{I}_3$, $\boldsymbol{\lambda} = \boldsymbol{\mu} = \boldsymbol{\alpha} = \mathrm{diag}(1, 1, 0.4)$, $\boldsymbol{\beta} = \mathrm{diag}(1e5, 1e5, 1)$ $\rho = 10$, $\boldsymbol{\Gamma} = 2\boldsymbol{I}_6$, $\kappa_1 = \kappa_2 = 0.1$, where $\boldsymbol{I}_N \in \mathbb{R}^{N\times N}$ is an

Authorized licensed use limited to: Korea Advanced Inst of Science & Tech - KAIST. Downloaded on June 02,2025 at 07:35:32 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c) (d)

Fig. 3. Simulation results: (a) tracking performance; (b) tracking error; (c) neural network weight estimation; (d) control input.

TABLE I
COMPARISON OF COST INDEXES IN SIMULATIONS

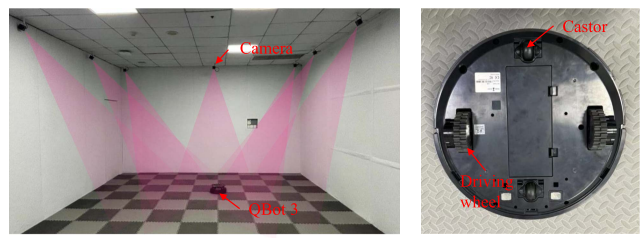
Controller	J_c		J_e	
	0-30 s	30-60 s	0-30 s	30-60 s
Proposed controller	8.837	6.592	0.402	0.002
BC	6.732	6.575	1.419	1.062
ADP-UUB	6.857	6.669	0.478	0.122

identity matrix. The parameters of BC are selected as $\xi_1 = 3$ and $\xi_2 = \xi_3 = 8$.

The simulation outcomes pertaining to the dynamic performance of the mobile robot are presented in Fig. 3. The motion trajectory of the center of mass in the $x - y$ plane is shown in Fig. 3(a). Meanwhile, the position tracking error and heading angle tracking error are depicted in Fig. 3(b). The results show that the proposed ADP-based fixed-time optimal tracking controller exhibits faster convergence and steady-state accuracy compared to the BC and ADP-UUB controllers. The learning process of neural network weight estimation is shown in Fig. 3(c), and it shows that all the weight estimations converge rapidly to different constants. The control input responses for the three controllers are shown in Fig. 3(d). To facilitate a more accurate comparison of the control performance between the three controllers, we define two specific performance indexes. Specifically, one is the control cost $J_c = \int_{t_0}^{t_f} \boldsymbol{u}^{\mathrm{T}}(\tau)\boldsymbol{\mathcal{R}}\boldsymbol{u}(\tau)\mathrm{d}\tau$ and the other is the error cost $J_e = \int_{t_0}^{t_f} \boldsymbol{z}^{\mathrm{T}}(\tau)\boldsymbol{\mathcal{Q}}\boldsymbol{z}(\tau)\mathrm{d}\tau$, where t_0 and t_f represent the initial and terminal moments, respectively. The performance indexes of the three controllers at different time periods are shown in Table I. The results show that compared to the other two controllers, the proposed controller requires more control cost in the first 30 s to achieve fast convergence of the tracking error. However, the error cost of the proposed controller is minimized for both time periods.

B. Experiment Results

To test the performance of the proposed method in a practical application, an experimental platform as shown in Fig. 4 is built. The OptiTrack system consists of multiple cameras, which can measure the position and attitude information of a robot in real time and transmit it to the host computer. The robot is QBot 3 manufactured by Quanser and is connected to the host computer via a wireless network. The controller is constructed in the host computer and downloaded to the QBot 3 via the wireless network. The QBot 3 can receive real-time state information from the host computer, thus realizing closed-loop control.



(a) OptiTrack system (b) Bottom view of the QBot 3

Fig. 4. Mobile robot experiment platform.

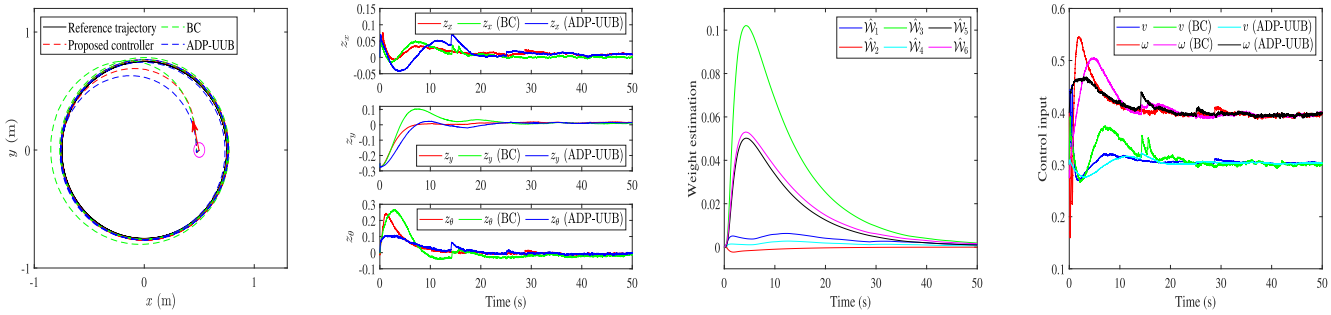
TABLE II
COMPARISON OF COST INDEXES IN EXPERIMENTS

Controller	J_c		J_e	
	0-25 s	25-50 s	0-25 s	25-50 s
Proposed controller	13.698	12.469	3.046	0.107
BC	14.362	12.326	4.859	0.134
ADP-UUB	13.504	12.461	3.065	0.089

The desired trajectory of the robot is defined as $[x_r(0), y_r(0), \theta_r(0)]^{\mathrm{T}} = [1.5, 0, \pi/2]^{\mathrm{T}}$, $v_r = 0.3$ m/s, $\omega_r = 0.4$ rad/s. The initial position of the robot is specified as $[x(0), y(0), \theta(0)]^{\mathrm{T}} = [0.5, 0, \pi/2]^{\mathrm{T}}$. The neural network basis functions are chosen in the same way as in the simulation. The parameters of the proposed optimal controller are chosen as follows: $\hat{\boldsymbol{\mathcal{W}}}(0) = [0, 0, 0, 0, 0, 0]^{\mathrm{T}}$, $p = 17$, $q = 19$, $\boldsymbol{\mathcal{R}} = 2\boldsymbol{I}_3$, $\boldsymbol{\mathcal{Q}} = 10\boldsymbol{I}_3$, $\boldsymbol{\lambda} = \boldsymbol{\mu} = \boldsymbol{\alpha} = \mathrm{diag}(0.1, 0.1, 0.2)$, $\boldsymbol{\beta} = \mathrm{diag}(200, 200, 0.3)$ $\rho = 1$, $\boldsymbol{\Gamma} = 2\boldsymbol{I}_6$, and $\kappa_1 = \kappa_2 = 0.1$. The parameters of BC are selected as $\xi_1 = \xi_2 = \xi_3 = 1.5$.

The experimental outcomes are depicted in Table II and Fig. 5. It is obvious that all closed-loop system signals remain bounded. According to the robot motion trajectory depicted in Fig. 5(a), the proposed controller and ADP-UUB can enable the robot to follow the desired trajectory rapidly, while the conventional BC cannot. Moreover, owing to the advantage of fixed-time stability, the proposed controller exhibits a faster convergence speed than that of the ADP-UUB. The tracking error is depicted by Fig. 5(b). Fig. 5(c) shows that the weight estimates of the neural network converge to a stable state after a period of learning process. The control signals are depicted by Fig. 5(d). Additionally, a comparison of the cost indexes of the three controllers is shown in Table II. According to the table, the error indexes of the

Authorized licensed use limited to: Korea Advanced Inst of Science & Tech - KAIST. Downloaded on June 02,2025 at 07:35:32 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c) (d)

Fig. 5. Experimental results: (a) tracking performance; (b) tracking error; (c) neural network weight estimation; (d) control input.

TABLE III
COST INDEXES FOR DIFFERENT CONTROL GAINS IN EXPERIMENTS

Control gains $\boldsymbol{\lambda} = \boldsymbol{\mu} = \boldsymbol{\alpha}$	J_c		J_e	
	0-25 s	25-50 s	0-25 s	25-50 s
diag(0.06, 0.06, 0.1)	13.946	12.447	4.090	0.085
diag(0.08, 0.08, 0.15)	14.223	12.408	4.004	0.084
diag(0.1, 0.1, 0.2)	13.698	12.469	3.046	0.107
diag(0.12, 0.12, 0.25)	14.317	12.550	3.109	0.119

proposed controller and ADP-UUB are better than the traditional BC for both time periods, which verifies the advantages of optimal control. On the other hand, comparing to the ADP-UUB, the proposed controller requires a bigger control cost to achieve fast tracking of the reference trajectory, which is in line with the practical significance. Finally, we have also conducted the experiment with the proposed controller under different control gains $\boldsymbol{\lambda}$, $\boldsymbol{\mu}$, and $\boldsymbol{\alpha}$ (the remaining parameters are consistent with previous experiments). The corresponding cost indexes are shown in Table III. It is clear that the control gains have a significant effect on the system's control performance, and suitable values need to be chosen to obtain the desired control performance.

V. CONCLUSION

By using the critic-only neural network ADP technique, an ADP-based fixed-time optimal tracking controller has been developed for wheeled mobile robots. The cost function is reconstructed using a critic neural network. Meanwhile, the adaptive control technique is utilized to design the weight update law without persistent or finite excitation conditions. Rigorous theoretical derivations demonstrate that the weight estimation error and the robot tracking error can converge to a neighborhood of zero in fixed time. Both the simulations and the experiments reveal that the proposed ADP-based fixed-time optimal tracking controller exhibits a superior convergence rate compared to the BC and ADP-UUB controllers. Based on the results of this study, we will study the optimal formation control for wheeled mobile robots in the future [39].

APPENDIX

This section presents two proofs for the inequalities used in the derivation of (22).

By the definition of $\hat{\boldsymbol{\mathcal{W}}}$, one has

$$\tilde{\mathcal{W}}_i\hat{\mathcal{W}}_i^3 = \tilde{\mathcal{W}}_i\left(\mathcal{W}_i^{\star 3} + 3\mathcal{W}_i^{\star}\tilde{\mathcal{W}}_i^2 - 3\mathcal{W}_i^{\star 2}\tilde{\mathcal{W}}_i - \tilde{\mathcal{W}}_i^3\right). \quad (35)$$

From Young's inequality (*Lemma 7* in [17]), one gets

$$\tilde{\mathcal{W}}_i\mathcal{W}_i^{\star 3} \le \frac{\mathcal{W}_i^{\star 4}}{12} + 3\mathcal{W}_i^{\star 2}\tilde{\mathcal{W}}_i^2 \quad (36)$$

$$3\tilde{\mathcal{W}}_i^{\star 3}\mathcal{W}_i \le \frac{9\varpi_i^{\frac{4}{3}}\tilde{\mathcal{W}}_i^4}{4} + \frac{3\mathcal{W}_i^{\star 4}}{4\varpi_i^4} \quad (37)$$

where $\varpi_i > 0$.

According to (35)–(37), the following inequality holds:

$$\tilde{\mathcal{W}}_i\hat{\mathcal{W}}_i^3 \le -\left(4 - 9\varpi_i^{\frac{4}{3}}\right)\left(\frac{1}{2}\tilde{\mathcal{W}}_i^2\right)^2 + \frac{\mathcal{W}_i^{\star 4}}{12} + \frac{3\mathcal{W}_i^{\star 4}}{4\varpi_i^4}. \quad (38)$$

Hence, by using *Lemma 2*, it can be obtained that

$$\begin{aligned}\kappa_2\left(\tilde{\boldsymbol{\mathcal{W}}}^{\mathrm{T}}\hat{\boldsymbol{\mathcal{W}}}\right)\hat{\boldsymbol{\mathcal{W}}}^{\mathrm{T}}\hat{\boldsymbol{\mathcal{W}}} \le & -\sum_{i=1}^{l}\left(4\kappa_2 - 9\kappa_2\varpi_i^{\frac{4}{3}}\right)\left(\frac{1}{2}\tilde{\mathcal{W}}_i^2\right)^2 \\ & + \sum_{i=1}^{l}\left(\frac{\kappa_2\mathcal{W}_i^{\star 4}}{12} + \frac{3\kappa_2\mathcal{W}_i^{\star 4}}{4\varpi_i^4}\right) \\ \le & -\phi_1\left(\frac{1}{2}\tilde{\boldsymbol{\mathcal{W}}}^{\mathrm{T}}\boldsymbol{\Gamma}^{-1}\tilde{\boldsymbol{\mathcal{W}}}\right)^2 \\ & + \sum_{i=1}^{l}\left(\frac{\kappa_2\mathcal{W}_i^{\star 4}}{12} + \frac{3\kappa_2\mathcal{W}_i^{\star 4}}{4\varpi_i^4}\right)\end{aligned} \quad (39)$$

where $\phi_1 = \min\{(4\kappa_2 - 9\kappa_2\varpi_1^{4/3})/[l(\vartheta_{\min}(\boldsymbol{\Gamma}^{-1}))^2], \ldots, (4\kappa_2 - 9\kappa_2\varpi_l^{4/3})/[l(\vartheta_{\min}(\boldsymbol{\Gamma}^{-1}))^2]\}$.

According to Young's inequality (*Lemma 7* in [17]), one gets

$$\kappa_1\tilde{\boldsymbol{\mathcal{W}}}^{\mathrm{T}}\hat{\boldsymbol{\mathcal{W}}} \le \frac{1}{2}\kappa_1\boldsymbol{\mathcal{W}}^{\star\mathrm{T}}\boldsymbol{\mathcal{W}}^{\star} - \frac{1}{2}\kappa_1\tilde{\boldsymbol{\mathcal{W}}}^{\mathrm{T}}\tilde{\boldsymbol{\mathcal{W}}}. \quad (40)$$

By *Lemma 1*, if make $\varrho = 1$, $\bar{\varrho} = (1/2)\kappa_1\tilde{\boldsymbol{\mathcal{W}}}^{\mathrm{T}}\tilde{\boldsymbol{\mathcal{W}}}$, $\mu = (p+q)/(2q)$, and $o = 1/\mu$, one has

$$\left(\frac{1}{2}\kappa_1\tilde{\boldsymbol{\mathcal{W}}}^{\mathrm{T}}\tilde{\boldsymbol{\mathcal{W}}}\right)^{\frac{p+q}{2q}} \le \frac{1}{2}\kappa_1\tilde{\boldsymbol{\mathcal{W}}}^{\mathrm{T}}\tilde{\boldsymbol{\mathcal{W}}} + \frac{q-p}{2q}\left(\frac{p+q}{2q}\right)^{\frac{p+q}{q-p}}. \quad (41)$$

Authorized licensed use limited to: Korea Advanced Inst of Science & Tech - KAIST. Downloaded on June 02,2025 at 07:35:32 UTC from IEEE Xplore. Restrictions apply.

Hence, one can obtain the inequality

$$\kappa_1 \tilde{\mathcal{W}}^{\mathrm{T}} \hat{\mathcal{W}} \leq \frac{1}{2}\kappa_1 \mathcal{W}^{\star\mathrm{T}} \mathcal{W}^{\star} - \delta_1 \left(\frac{1}{2}\tilde{\mathcal{W}}^{\mathrm{T}} \boldsymbol{\Gamma}^{-1} \tilde{\mathcal{W}}\right)^{\frac{p+q}{2q}} + \frac{q-p}{2q}\left(\frac{p+q}{2q}\right)^{\frac{p+q}{q-p}} \tag{42}$$

where $\delta_1 = (\kappa_1/\vartheta_{\min}(\boldsymbol{\Gamma}^{-1}))^{(p+q/2q)}$, and $\vartheta_{\min}(\boldsymbol{\Gamma}^{-1})$ represents the maximum eigenvalue of $\boldsymbol{\Gamma}^{-1}$.

References

[1] Z. Li, W. Gao, C. Goh, M. Yuan, E. K. Teoh, and Q. Ren, "Asymptotic stabilization of nonholonomic robots leveraging singularity," *IEEE Robot. Autom. Lett.*, vol. 4, no. 1, pp. 41–48, Jan. 2019.

[2] S. J. Yoo, "Adaptive tracking control for a class of wheeled mobile robots with unknown skidding and slipping," *IET Control Theory Appl.*, vol. 4, no. 10, pp. 2109–2119, Oct. 2010.

[3] H. Gao, X. Song, L. Ding, K. Xia, N. Li, and Z. Deng, "Adaptive motion control of wheeled mobile robot with unknown slippage," *Int. J. Control*, vol. 87, no. 8, pp. 1513–1522, Feb. 2014.

[4] C. Chen, H. Gao, L. Ding, W. Li, H. Yu, and Z. Deng, "Trajectory tracking control of WMRs with lateral and longitudinal slippage based on active disturbance rejection control," *Robot. Auton. Syst.*, vol. 107, pp. 236–245, Sep. 2018.

[5] M. Mera, H. Ros, and E. A. Martnez, "A sliding-mode based controller for trajectory tracking of perturbed unicycle mobile robots," *Control Eng. Pract.*, vol. 102, Sep. 2020, Art. no. 104548.

[6] P. Rochel, H. Ros, M. Mera, and A. Dzul, "Trajectory tracking for uncertain unicycle mobile robots: A super-twisting approach," *Control Eng. Pract.*, vol. 122, May 2022, Art. no. 105078.

[7] H. Yang, X. Fan, Y. Xia, and C. Hua, "Robust tracking control for wheeled mobile robot based on extended state observer," *Adv. Robot.*, vol. 30, no. 1, pp. 68–78, Jan. 2016.

[8] H. Wang, P. X. Liu, X. Zhao, and X. Liu, "Adaptive fuzzy finite-time control of nonlinear systems with actuator faults," *IEEE Trans. Cybern.*, vol. 50, no. 5, pp. 1786–1797, May 2020.

[9] H. Wang, W. Liu, and M. Tong, "Adaptive fuzzy fast finite-time output-feedback tracking control for switched nonlinear systems with full-state constraints," *IEEE Trans. Fuzzy Syst.*, vol. 32, no. 3, pp. 958–968, Mar. 2024.

[10] J. Wang, C. Wang, C. L. P. Chen, Z. Liu, and C. Zhang, "Fast finite-time event-triggered consensus control for uncertain nonlinear multiagent systems with full-state constraints," *IEEE Trans. Circuits Syst. I., Reg. Papers*, vol. 70, no. 3, pp. 1361–1370, Mar. 2023.

[11] S. Li and Y.-P. Tian, "Finite-time stability of cascaded time-varying systems," *Int. J. Control*, vol. 80, no. 4, pp. 646–657, Apr. 2007.

[12] D. Wu, Y. Cheng, H. Du, W. Zhu, and M. Zhu, "Finite-time output feedback tracking control for a nonholonomic wheeled mobile robot," *Aerosp. Sci. Technol.*, vol. 78, pp. 574–579, Jul. 2018.

[13] H. Wang, K. Xu, and H. Zhang, "Adaptive finite-time tracking control of nonlinear systems with dynamics uncertainties," *IEEE Trans. Autom. Control*, vol. 68, no. 9, pp. 5737–5744, Sep. 2023.

[14] Y.-X. Li, Z. Hou, W.-W. Che, and Z.-G. Wu, "Event-based design of finite-time adaptive control of uncertain nonlinear systems," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 8, pp. 3804–3813, Aug. 2022.

[15] C. P. Vo and K. K. Ahn, "An adaptive finite-time force-sensorless tracking control scheme for pneumatic muscle actuators by an optimal force estimation," *IEEE Robot. Autom. Lett.*, vol. 7, no. 2, pp. 1542–1549, Apr. 2022.

[16] A. Polyakov, "Nonlinear feedback design for fixed-time stabilization of linear control systems," *IEEE Trans. Autom. Control*, vol. 57, no. 8, pp. 2106–2110, Aug. 2012.

[17] H. Wang, J. Ma, X. Zhao, B. Niu, M. Chen, and W. Wang, "Adaptive fuzzy fixed-time control for high-order nonlinear systems with sensor and actuator faults," *IEEE Trans. Fuzzy Syst.*, vol. 31, no. 8, pp. 2658–2668, Aug. 2023.

[18] H. Wang and L. Shen, "Adaptive fuzzy fixed-time tracking control for nonlinear systems with time-varying full-state constraints and actuator hysteresis," *IEEE Trans. Fuzzy Syst.*, vol. 31, no. 4, pp. 1352–1361, Apr. 2023.

[19] C. Wang, Q. Guo, J. Wang, Z. Liu, and C. L. P. Chen, "Fixed-time fuzzy control for uncertain nonlinear systems with prescribed performance and event-triggered communication," *IEEE Trans. Circuits Syst. I., Reg. Papers*, vol. 71, no. 5, pp. 2362–2371, May 7, 2024.

[20] M. Chen, H. Wang, and X. Liu, "Adaptive practical fixed-time tracking control with prescribed boundary constraints," *IEEE Trans. Circuits Syst. I., Reg. Papers*, vol. 68, no. 4, pp. 1716–1726, Apr. 2021.

[21] Y. Zhang and F. Wang, "Observer-based fixed-time neural control for a class of nonlinear systems," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 7, pp. 2892–2902, Jul. 2022.

[22] J. Zhang, F. Xu, X. Liu, S. Gu, and H. Geng, "Fixed-time dynamic surface control for pneumatic manipulator system with unknown disturbances," *IEEE Robot. Autom. Lett.*, vol. 7, no. 4, pp. 10890–10897, Oct. 2022.

[23] C. Wang, H. Zhan, Q. Guo, and T. Li, "Distributed neural fixed-time consensus control of uncertain multiple Euler-Lagrange systems with event-triggered mechanism," *IEEE/ASME Trans. Mechatron.*, early access: Jun., 25, 2024, doi: 10.1109/TMECH.2024.3410299.

[24] Z. Lv, Y. Wu, X.-M. Sun, and Q.-G. Wang, "Fixed-time control for a quadrotor with a cable-suspended load," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 11, pp. 21932–21943, Nov. 2022.

[25] H. Dong, X. Zhao, Q. Hu, H. Yang, and P. Qi, "Learning-based attitude tracking control with high-performance parameter estimation," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 58, no. 3, pp. 2218–2230, Jun. 2022.

[26] R. E. Bellman, *Dynamic Programming*. Princeton, NJ, USA: Princeton Univ. Press, 1957.

[27] D. Liu, D. Wang, F.-Y. Wang, H. Li, and X. Yang, "Neural-network-based online HJB solution for optimal robust guaranteed cost control of continuous-time uncertain nonlinear systems," *IEEE Trans. Cybern.*, vol. 44, no. 12, pp. 2834–2847, Dec. 2014.

[28] D. Wang and C. Mu, "Adaptive-critic-based robust trajectory tracking of uncertain dynamics and its application to a spring-mass-damper system," *IEEE Trans. Ind. Electron.*, vol. 65, no. 1, pp. 654–663, Jan. 2018.

[29] C. Mu, Y. Zhang, Z. Gao, and C. Sun, "ADP-based robust tracking control for a class of nonlinear systems with unmatched uncertainties," *IEEE Trans. Syst. Man Cybern. Syst.*, vol. 50, no. 11, pp. 4056–4067, Nov. 2020.

[30] G. Wen, S. S. Ge, and F. Tu, "Optimized backstepping for tracking control of strict-feedback systems," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 8, pp. 3850–3862, Aug. 2018.

[31] Y. Li, K. Li, and S. Tong, "Reinforcement learning-based adaptive finite-time performance constraint control for nonlinear systems," *IEEE Trans. Syst. Man Cybern. Syst.*, vol. 54, no. 2, pp. 1335–1344, Feb. 2024.

[32] H. Dong, X. Zhao, and B. Luo, "Optimal tracking control for uncertain nonlinear systems with prescribed performance via critic-only ADP," *IEEE Trans. Syst. Man Cybern. Syst.*, vol. 52, no. 1, pp. 561–573, Jan. 2022.

[33] B. Xiao, H. Zhang, Z. Chen, and L. Cao, "Fixed-time fault-tolerant optimal attitude control of spacecraft with performance constraint via reinforcement learning," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 59, no. 6, pp. 7715–7724, Dec. 2023.

[34] R. Kamalapurkar, H. S. D. Bhasin, and W. E. Dixon, "Approximate optimal trajectory tracking for continuous-time nonlinear systems," *Automatica*, vol. 51, pp. 40–48, Jan. 2015.

[35] C.-Y. Chen, T.-H. S. Li, Y.-C. Yeh, and C.-C. Chang, "Design and implementation of an adaptive sliding-mode dynamic controller for wheeled mobile robots," *Mechatron.*, vol. 19, no. 2, pp. 156–166, Mar. 2009.

[36] J. Zhang, S. Li, and Z. Xiang, "Adaptive fuzzy output feedback event-triggered control for a class of switched nonlinear systems with sensor failures," *IEEE Trans. Circuits Syst. I., Reg. Papers*, vol. 67, no. 12, pp. 5336–5346, Dec. 2020.

[37] D. Ba, Y. Li, and S. Tong, "Fixed-time adaptive neural tracking control for a class of uncertain nonstrict nonlinear systems," *Neurocomputing*, vol. 363, pp. 273–280, Oct. 2019.

[38] S. Yang, F. Yu, H. Liu, H. Ma, and H. Zhang, "Adaptive-dynamic-programming-based robust control for a quadrotor UAV with external disturbances and parameter uncertainties," *Appl. Sci.*, vol. 13, no. 23, Nov. 2023, Art. no. 12672.

[39] W. Liu, X. Wang, and S. Li, "Formation control for leader-follower wheeled mobile robots based on embedded control technique," *IEEE Trans. Control Syst. Technol.*, vol. 31, no. 1, pp. 265–280, Jan. 2023.

Authorized licensed use limited to: Korea Advanced Inst of Science & Tech - KAIST. Downloaded on June 02,2025 at 07:35:32 UTC from IEEE Xplore. Restrictions apply.